Fe-Ni-Poly(heptazine imides) as electrocatalysts for zinc-air batteries – Influence of metal species and content

Franz Jacobi, Marius Hermesdorf, Rudolf Frank Baumgart, Sri Rezeki, Desireé Leistenschneider, Center for Energy and Environmental Chemistry, Philosophenweg 7a Friedrich-Schiller-Universität, 07743 Jena/Germany

Poly(heptazine imide)s (PHI) are polymeric 2D-materials of the carbon nitride family.[1] Their distinguishing feature is their negatively charged imide bridges, which are counterbalanced by positively charged ions. Those ions further grant the PHI its high crystallinity and influence their stacking behavior. PHI with imide-K^+ sites (K-PHI) are the most investigated representative of this material class and can be obtained by ionothermal synthesis using Liebig's melon - a carbon nitride - and KSCN.[2] Their metal cations can be exchanged by other positively charged species, marking PHIs as versatile platforms for catalyst design as the cations influence material properties such as Lewis acidity and their electronic and ionic properties, comparable to zeolites.[3] Contrarily to electrically insulating zeolites however, PHIs as semiconductors can be designed for electrical applications as well. Previous research already investigated Co-PHI as an electrocatalyst for the oxygen evolution reaction for the water splitting reaction.[4] Furthermore, hybridization of PHIs with highly conductive materials like carbon paves the way towards efficient application of this material towards the field of electrocatalysis.[5]

Zinc-air batteries (ZABs) are energy storage systems that are focus of current research for their high energy density use of abundant materials and potential application for grid energy storage systems. While zinc serves as the anode, an electrocatalyst immobilized at a gas diffusion layer (GDL) catalyzes the reduction of oxygen from the air to hydroxide and separates the electrolyte from the oxygen cathode. Commonly, noble metal catalysts like Pt are used for catalyzing the oxygen reduction reaction (ORR), impeding affordable large-scale production.[6] Furthermore, improving the rechargeability relying on the oxygen evolution reaction (OER) is necessary for long-term applications.

This work investigates the fabrication of transition metal doped PHIs, namely Fe-PHI and Ni-PHI, for application as ORR and OER electrocatalysts respectively. Whereas promising ORR activities are observable for Fe-PHI, Ni-PHI reaches OER activities

comparable to the benchmark catalyst RuO_2. For that reason, bifunctional FeNi-PHI was synthesized, a form of mixed transition metal PHI, which has not been applied as electrocatalyst to the best of our knowledge, yet. High resolution transmission electron microscopy coupled with energy dispersive X-ray spectroscopy of FeNi-PHI indicate a homogenous distribution of the elements within this highly ordered material. In this context, the PHI materials are applied as catalysts at the GDL in a ZAB setup. Here, the charging performance of the noble metal reference material is surpassed by the FeNi-PHI catalyst, and a charge-discharge potential gap of 0.81 V shows the bifunctionality of the material. Additionally, over the course of 250 charge-discharge cycles the system's performance stays steady, resulting in good long-term stability and rechargeability.[7]

[1] V. W. Lau and B. V. Lotsch, *Adv. Energy Mater.*, **2022**, 12, 2101078.

[2] A. Gouder, F. Podjaski, A. Jiménez-Solano, J. Kröger, Y. Wang and B. V. Lotsch, *Energy Environ. Sci.*, **2023**, 16, 1520–1530.

[3] A. Savateev, S. Pronkin, M. G. Willinger, M. Antonietti and D. Dontsova, *Chem.-Asian J.*, **2017**, 12, 1517–1522.

[4] M. Ye, S. Li, X. Zhao, N. V. Tarakina, C. Teutloff, W. Y. Chow, R. Bittl and A. Thomas, *Adv. Mater.*, **2020**, 32, 1903942.

[5] C. M. Pelicano, J. Li, M. Cabrero-Antonino, I. F. Silva, L. Peng, N. V. Tarakina, S. Navalón, H. García and M. Antonietti, *J. Mater. Chem. A*, **2024**, 12, 475–482.

[6] M. Ye, J. Wang, L. Zhan, M. Wang, W. Yan, Y. Wang, K. Yang, H. Liu, Y. Tan, W. Wang, C. Chen and K. Wu, *ACS Appl. Energy Mater.*, **2024**, acsaem.3c02810.

[7] F. Jacobi, M. Hermesdorf, R. F. Baumgart, S. Rezeki, D. Leistenschneider, "Fe-Ni-Poly(heptazine imides) as electrocatalysts for zinc-air batteries – Influence of metal species and content", in preparation.